

Simulation-based verification of **bfpwr**

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This vignette summarizes the simulation-based verification checks for **bfpwr**. The report focuses on the verification settings and results: the simulated designs, analysis priors, evidence thresholds, sampling schedules, and the size of any package calculation differences relative to the simulation targets.

1 z-Test

Simulated Design Priors

Table 1: Simulated design priors for the z-Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	USD
Z1	Point(0)	10 to 500 by 10	10000	1
Z2	Point(0.2)	10 to 500 by 10	10000	1
Z3	Point(0.5)	10 to 500 by 10	10000	1
Z4	Normal(0.5, 0.1)	10 to 500 by 10	10000	1
Z5	Normal(0.2, 0.3)	10 to 500 by 10	10000	1
Z6	Normal(0.2, 0.5)	10 to 500 by 10	10000	1
Z7	Normal(0, 0.5)	10 to 500 by 10	10000	1
Z8	Point(-0.3)	10 to 500 by 10	10000	1
Z9	Point(0)	10 to 500 by 10	10000	50
Z10	Point(10)	10 to 500 by 10	10000	50
Z11	Normal(10, 20)	10 to 500 by 10	10000	50
Z12	Point(0)	10 to 10000 by 10	10000	1
Z13	Point(0.2)	10 to 10000 by 10	10000	1
Z14	Normal(0.2, 0.1)	10 to 10000 by 10	10000	1
Z15	Normal(0.2, 0.5)	10 to 10000 by 10	10000	1

Analysis Priors

Table 2: Analysis priors used for the z-Test verification run. Designs lists the matching short IDs from the design-prior table.

Function	H1	H0	Designs
bf01	Normal(0, 0.707)	Point(0)	Z1-Z8, Z12-Z15
bf01	Normal(0, 20)	Point(0)	Z9-Z11
bf01	Normal(0.2, 0.3)	Point(0)	Z1-Z8, Z12-Z15

bf01	Normal(0.2, 0.707)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Normal(10, 20)	Point(0)	Z9-Z11
bf01	Point(0.2)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0)	Z1-Z8, Z12-Z15
bf01	Point(0.5)	Point(0.2)	Z2-Z6, Z13-Z15
bf01	Point(10)	Point(0)	Z9-Z11
dirbf01	Normal ₊ (0, 0.707)	Point(0)	Z1-Z8, Z12-Z15
dirbf01	Normal ₊ (0, 20)	Point(0)	Z9-Z11
dirbf01	Normal ₊ (0.2, 0.707)	Point(0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment(0.354)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(0.354)	Point(0.2)	Z2-Z6, Z13-Z15
nmbf01	Moment(0.707)	Point(0)	Z1-Z8, Z12-Z15
nmbf01	Moment(7.071)	Point(0)	Z9-Z11

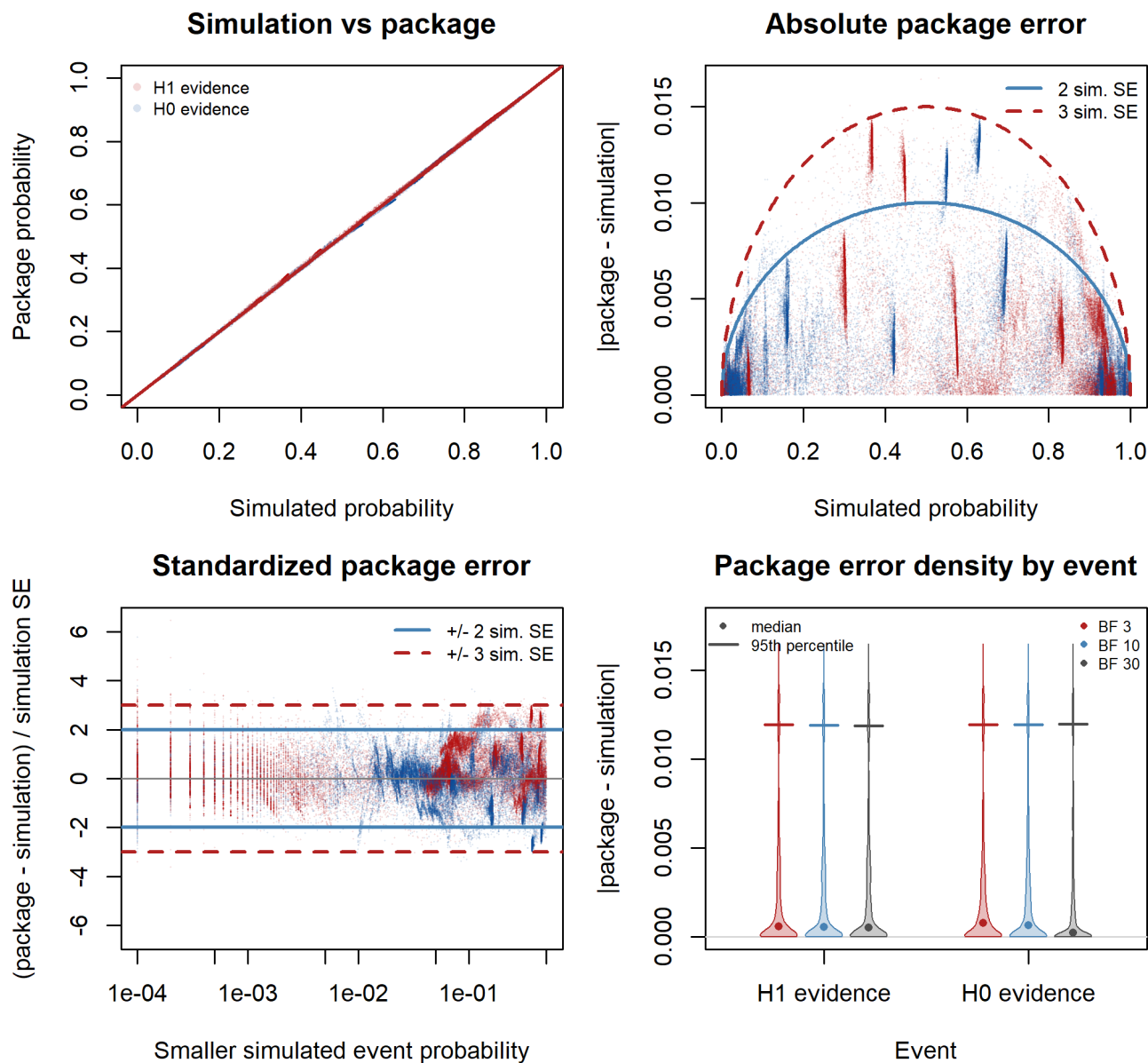
1.1 Fixed-Design

Table 3: Fixed-design fixture setups for z-Test analyses.

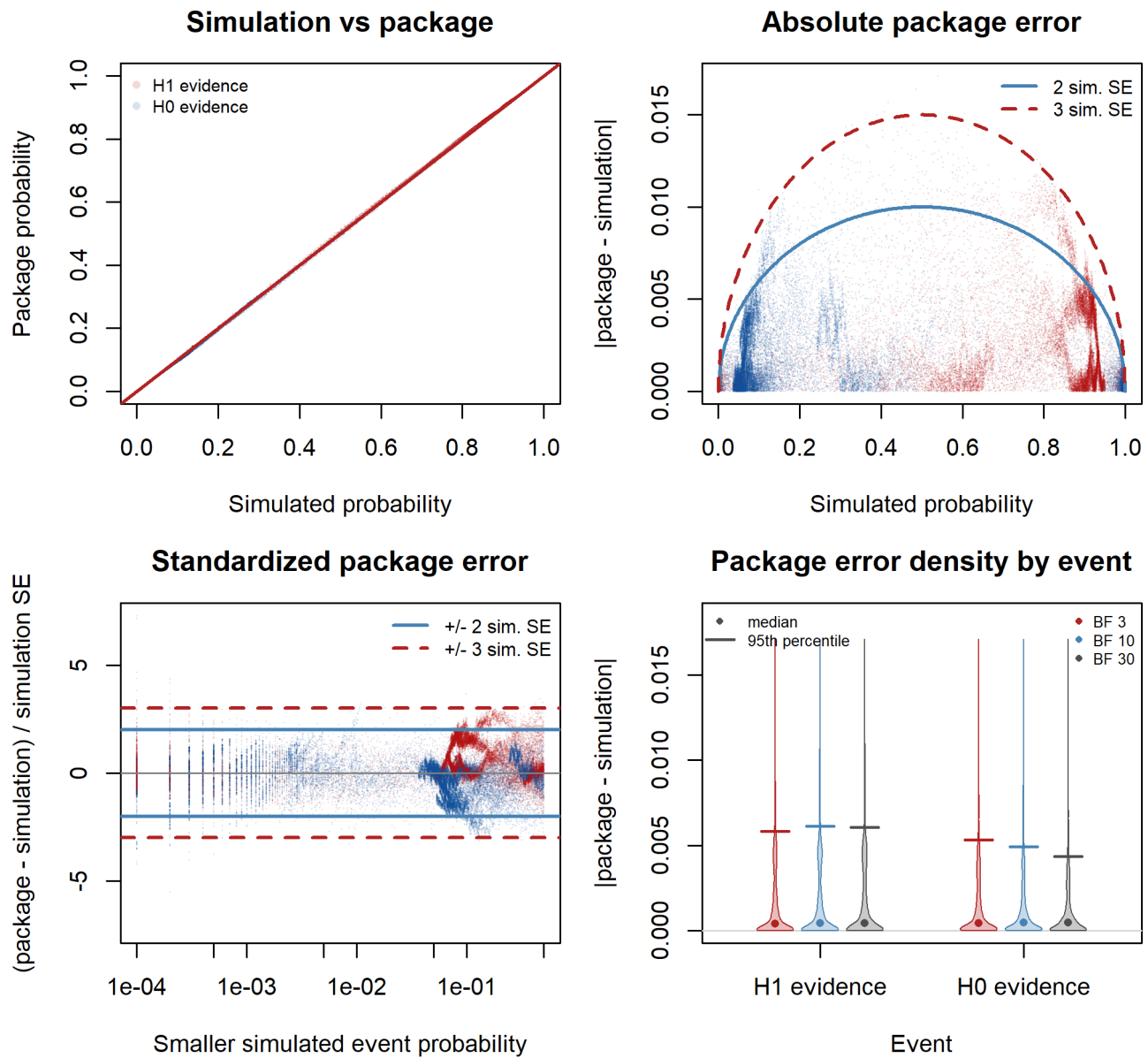
Function	Analysis priors	Evidence thresholds
bf01	73	BF 3, 10, 30
nmbf01	35	BF 3, 10, 30
dirbf01	23	BF 3, 10, 30

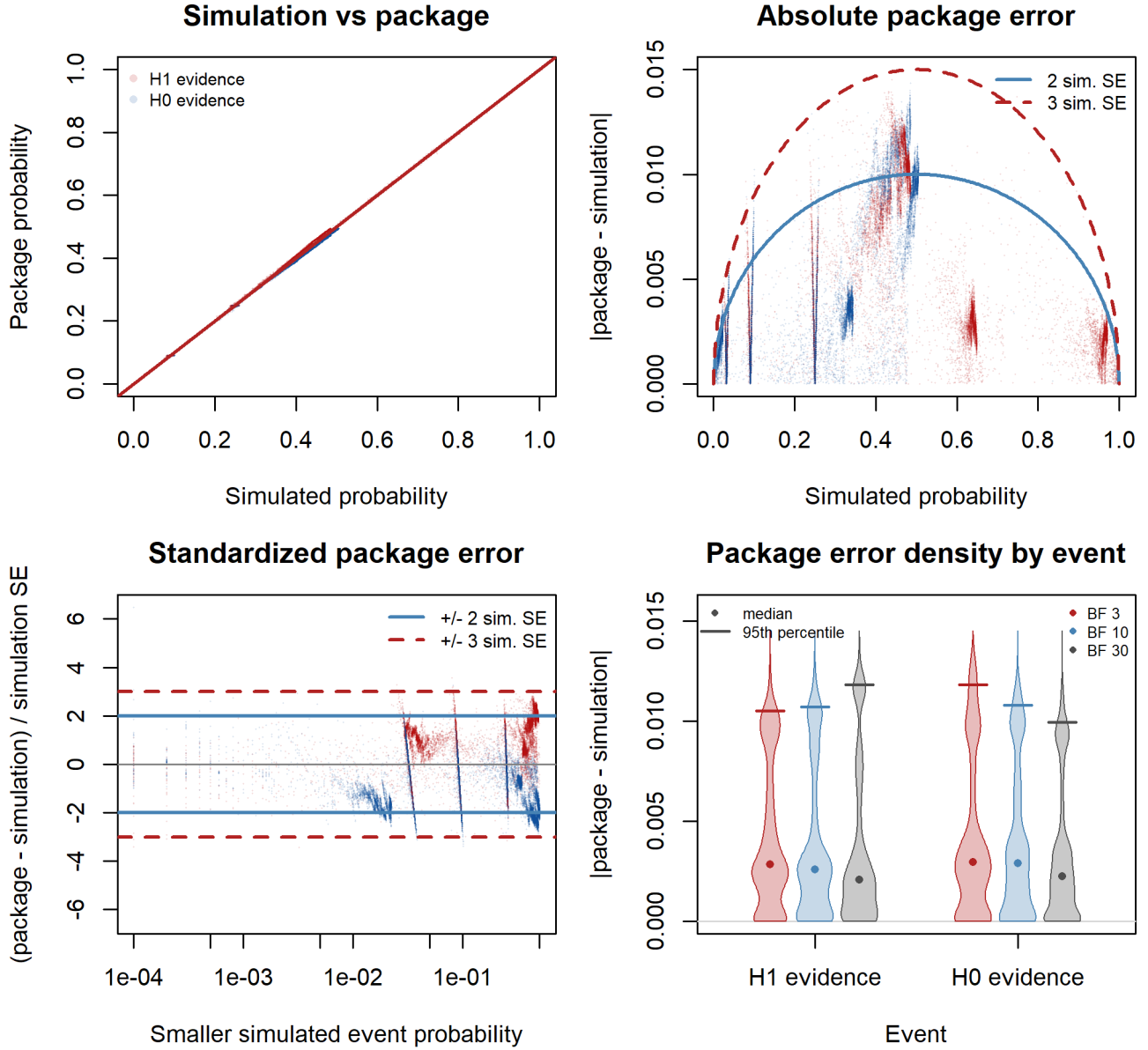
The fixed-design plots validate package probability calculations against the simulated event frequencies. Cached diagnostic rows are shown; for computationally expensive reference families, the plotted cache can be a stratified subset of the full fixture summary. The panels compare simulated and package probabilities, show absolute and standardized package error, and show package-error densities by event.

bf01 fixed-design fixtures



nmbf01 fixed-design fixtures



dirbf01 fixed-design fixtures**1.2 Sequential-Design**

Sequential fixture schedules use looks every 10 observations: one family starts at $n = 20$ and includes 2–20 looks; the stress schedules start at $n = 10$ and include 50, 100, or 200 looks. The 100- and 200-look schedules are compatible with the longer design sequence only.

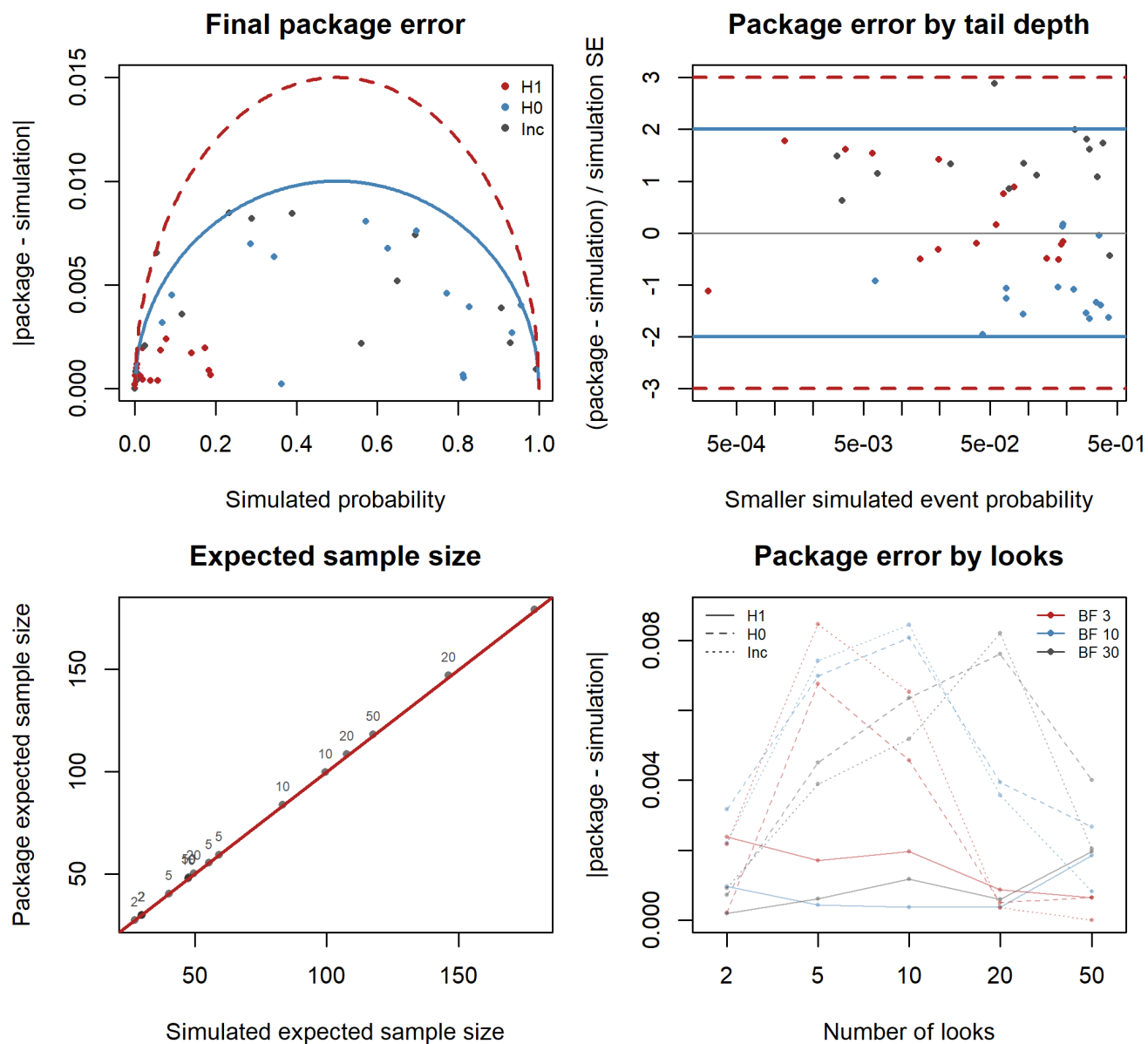
Table 4: Sequential-design fixture setups for z-Test analyses.

Function	Analysis priors	Evidence thresholds	Look sequence
bff01	73	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
nmbff01	35	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
dirbf01	23	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)

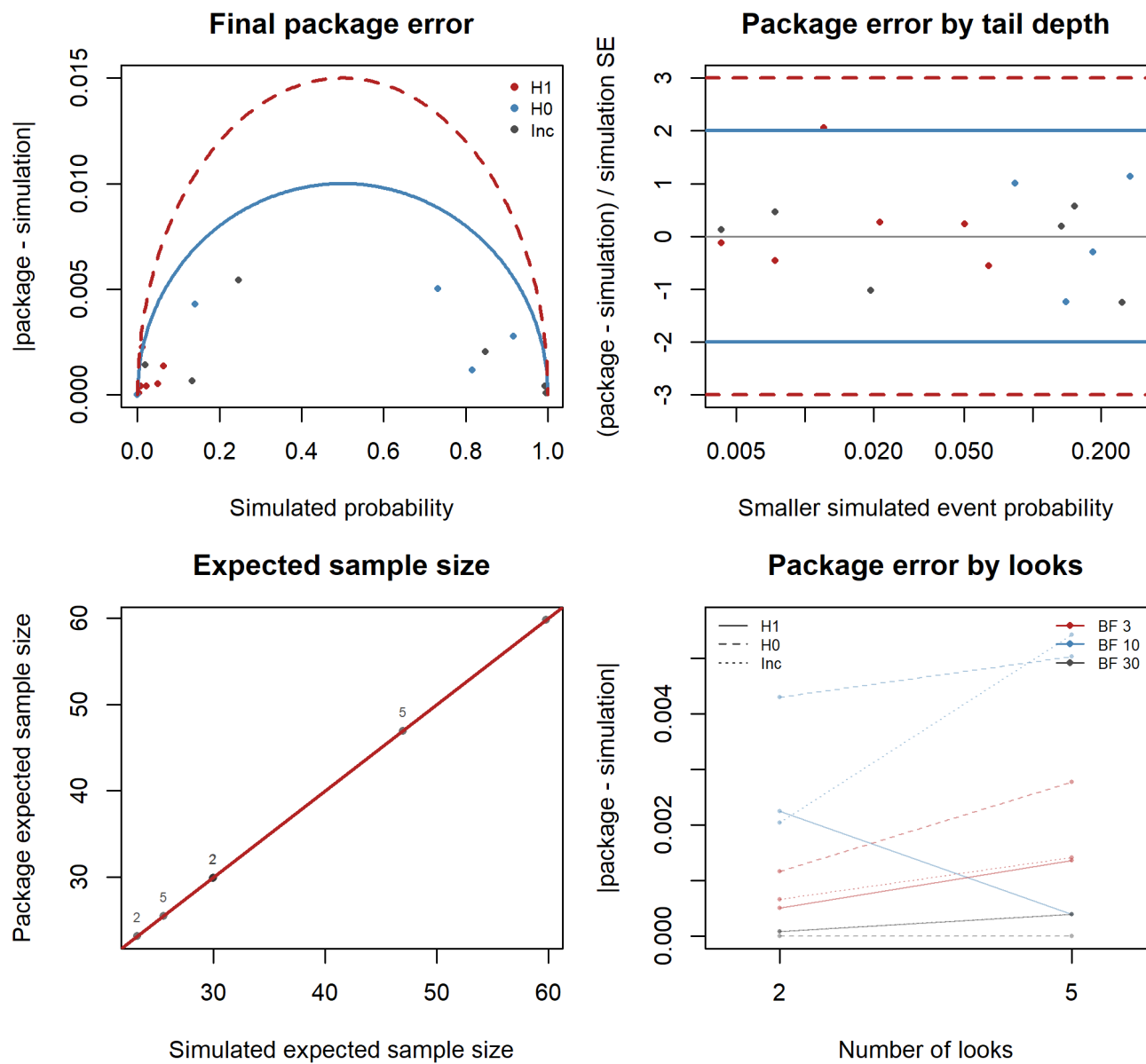
Sequential-design plots validate package probability calculations against selected simulated fixture sum-

maries. The panels show final package error, package error by tail depth, expected sample size agreement, and design-level package-error traces by number of looks. Expected sample size points use the same selected fixture rows as the probability checks and are not aggregated. Package calculations are checked at 2, 5, 10, 20, and 50 looks for normal and directional z -test priors. Moment-prior checks currently cover 2 and 5 looks; the remaining sequential schedules are retained in the settings table and automated fixtures.

bf01 sequential-design fixtures



nmbf01 sequential-design fixtures



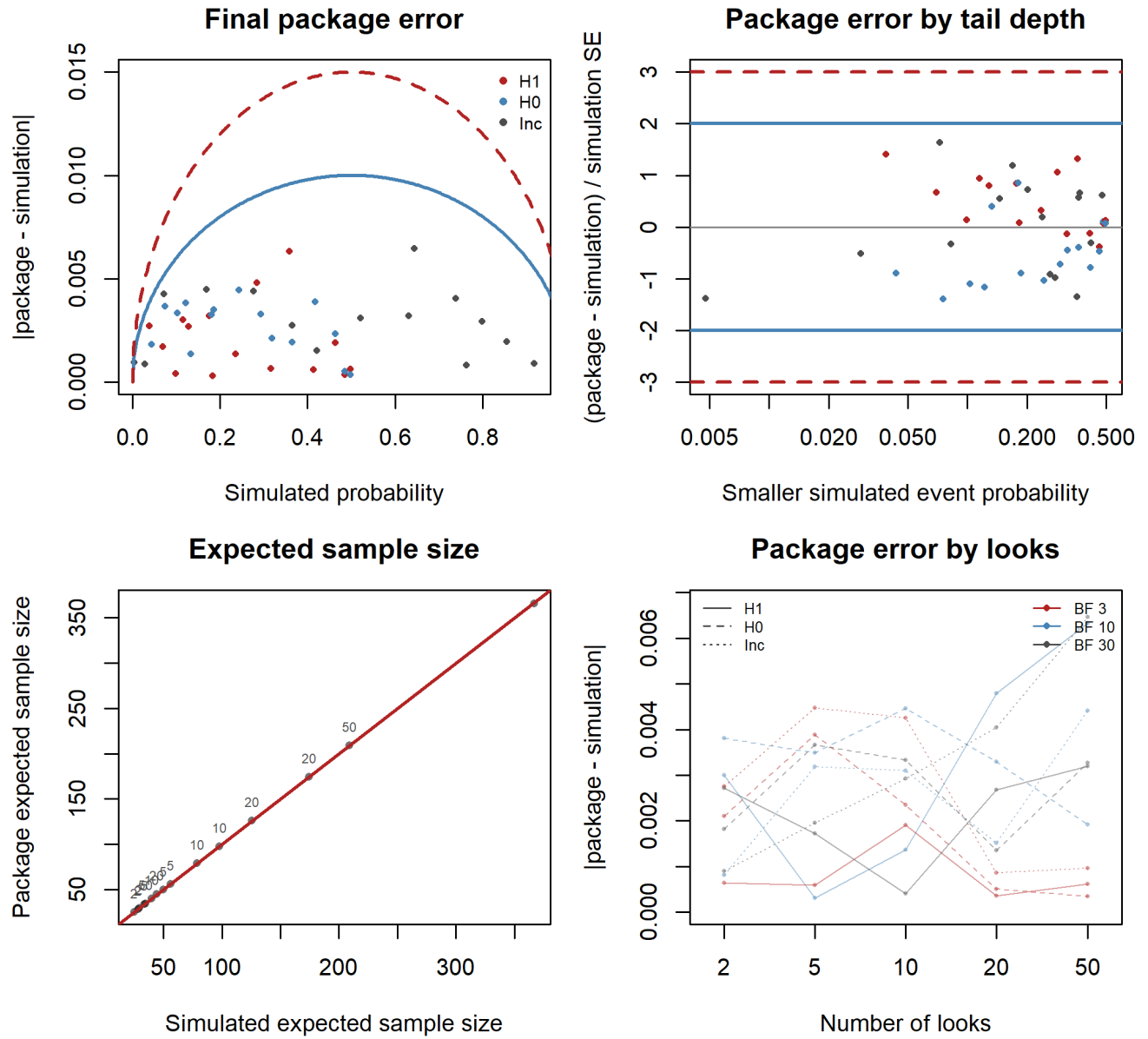
dirbf01 sequential-design fixtures**2 *t*-Test****Simulated Design Priors****Analysis Priors**

Table 6: Analysis priors used for the *t*-Test verification run. Designs lists the matching short IDs from the design-prior table.

Function	H1	H0	Designs
tbfb01	Cauchy ₋ (0, 0.707)	Point(0)	T1-T16
tbfb01	Cauchy ₊ (0, 0.707)	Point(0)	T1-T16
tbfb01	Cauchy(0, 0.35)	Point(0)	T1-T16

tbf01	Cauchy(0, 0.707)	Point(0)	T1-T16
tbf01	Cauchy(0, 1)	Point(0)	T1-T16
tbf01	StudentT ₊ (3, 0.5, 0.1)	Point(0)	T2-T4, T6, T8, T11, T14-T16
tbf01	StudentT ₊ (3, 0.5, 0.35)	Point(0.2)	T11, T14-T16
tbf01	StudentT ₊ (30, 0.5, 0.35)	Point(0.2)	T1, T5, T7, T10, T13
tbf01	StudentT(3, 0, 0.707)	Point(0)	T1-T16
tbf01	StudentT(30, 0, 0.707)	Point(0)	T1-T16

2.1 Fixed-Design

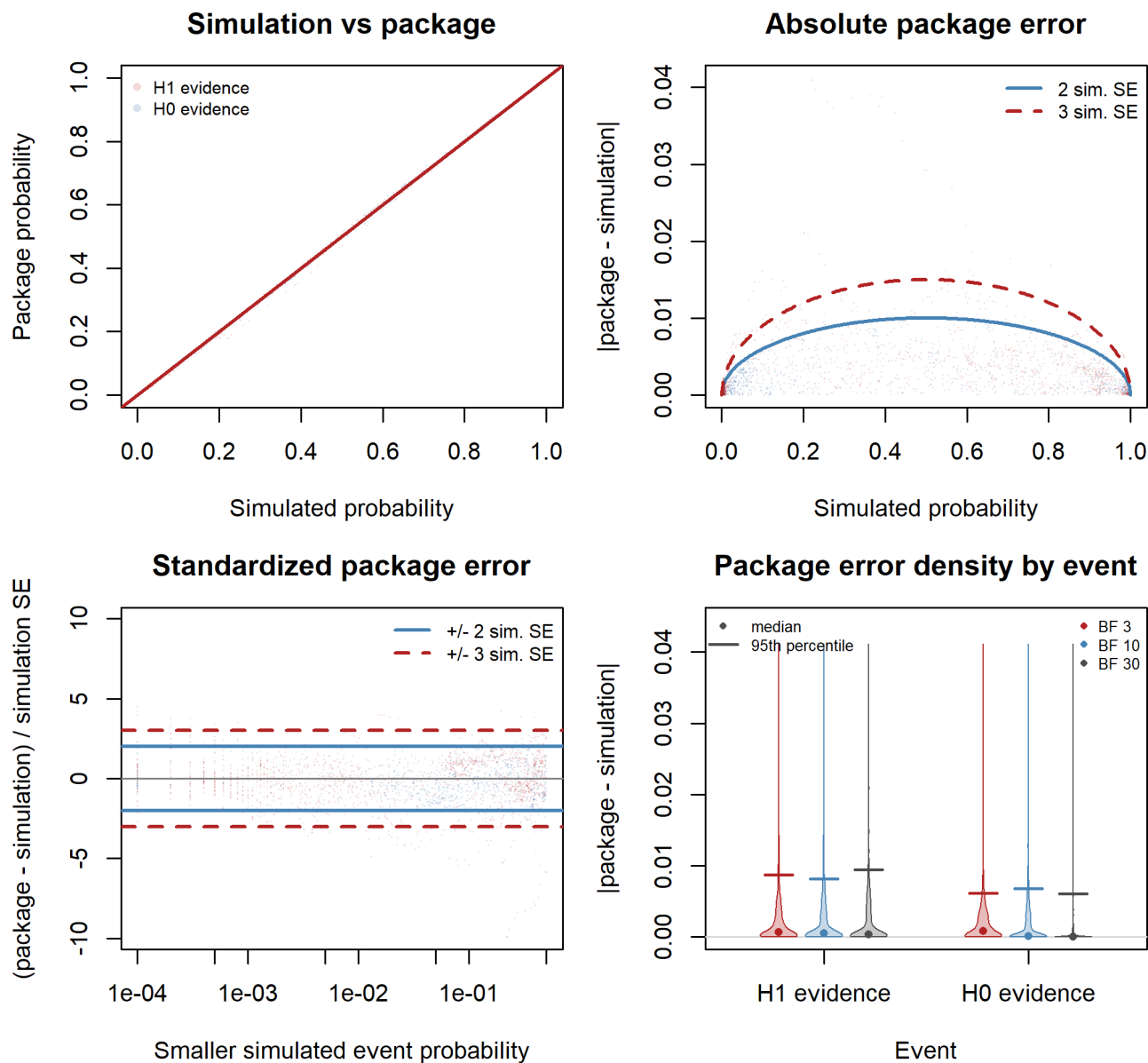
Table 7: Fixed-design fixture setups for t-Test analyses.

Function	Analysis priors	Evidence thresholds
tbf01	130	BF 3, 10, 30

The fixed-design plots validate package probability calculations against the simulated event frequencies. Cached diagnostic rows are shown; for computationally expensive reference families, the plotted cache can be a stratified subset of the full fixture summary. The panels compare simulated and package probabilities, show absolute and standardized package error, and show package-error densities by event.

Table 5: Simulated design priors for the t-Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions	Sampling	n2/n1
T1	Point(0)	10 to 500 by 10	10000	two-sample	1
T2	Point(0.5)	10 to 500 by 10	10000	two-sample	1
T3	Normal(0.5, 0.1)	10 to 500 by 10	10000	two-sample	1
T4	Normal(0.5, 0.5)	10 to 500 by 10	10000	two-sample	1
T5	Point(0)	10 to 500 by 10	10000	one-sample	
T6	Point(0.5)	10 to 500 by 10	10000	one-sample	
T7	Point(0)	10 to 500 by 10	10000	paired	
T8	Point(0.5)	10 to 500 by 10	10000	paired	
T9	Point(-0.4)	10 to 500 by 10	10000	two-sample	1
T10	Point(0)	10 to 500 by 10	10000	two-sample	1.1
T11	Point(0.35)	10 to 500 by 10	10000	two-sample	1.1
T12	Normal(0, 0.5)	10 to 500 by 10	10000	two-sample	1
T13	Point(0)	10 to 10000 by 10	10000	two-sample	1
T14	Point(0.2)	10 to 10000 by 10	10000	two-sample	1
T15	Normal(0.2, 0.1)	10 to 10000 by 10	10000	two-sample	1
T16	Normal(0.2, 0.5)	10 to 10000 by 10	10000	two-sample	1

tbf01 fixed-design fixtures**2.2 Sequential-Design**

Sequential fixture schedules use looks every 10 observations: one family starts at $n = 20$ and includes 2–20 looks; the stress schedules start at $n = 10$ and include 50, 100, or 200 looks. The 100- and 200-look schedules are compatible with the longer design sequence only.

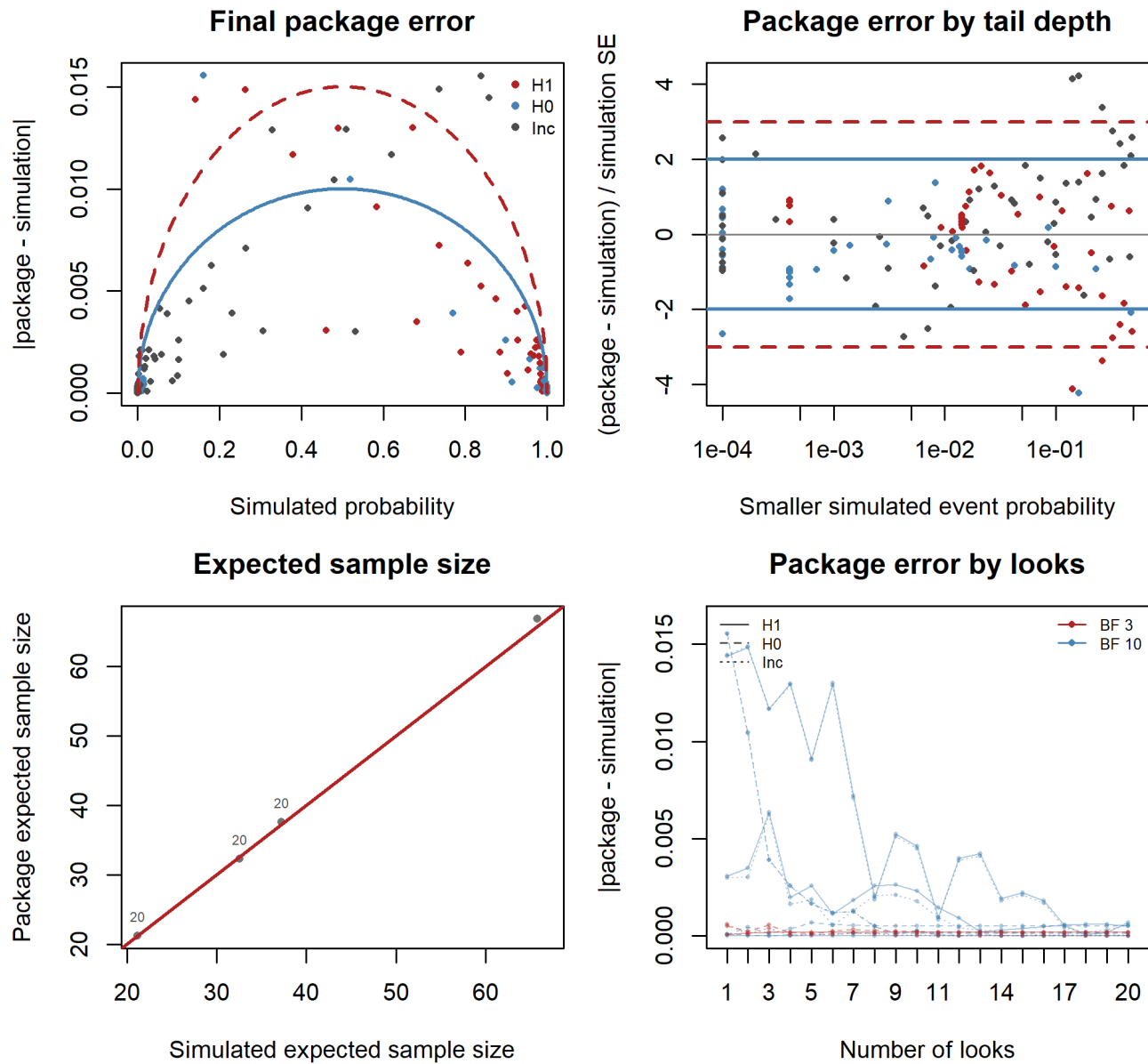
Table 8: Sequential-design fixture setups for t-Test analyses.

Function	Analysis priors	Evidence thresholds	Look sequence
tbf01	130	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)

Sequential-design plots validate selected one-sided t -test fixture summaries against strict `ptbf01seq()` calculations at 20 looks. Shifted-null unequal two-sample, paired, and 50/100/200-look stress schedules

are retained in the automated fixture tests as manifest, deterministic-summary, and row-presence checks until the sequential t -test reference path is improved for those settings.

tbf01 sequential-design fixtures



3 Binomial Test

Simulated Design Priors

Table 9: Simulated design priors for the Binomial Test verification run. The Design column gives short IDs used by the analysis-prior table.

Design	Design Prior	Sequence	Repetitions
B1	Point(0.5)	10 to 500 by 10	10000
B2	Point(0.6)	10 to 500 by 10	10000
B3	Point(0.2)	10 to 500 by 10	10000

B4	Point(0.75)	10 to 500 by 10	10000
B5	Beta(60, 40)	10 to 500 by 10	10000
B6	Beta(1, 1) on [0.5, 1]	10 to 500 by 10	10000
B7	Beta(1, 1) on [0, 0.5]	10 to 500 by 10	10000
B8	Point(0.5)	10 to 10000 by 10	10000
B9	Point(0.55)	10 to 10000 by 10	10000

Analysis Priors

Table 10: Analysis priors used for the Binomial Test verification run. Designs lists the matching short IDs from the design-prior table.

Function	H1	H0	Designs
binbf01 (direction)	Beta ₊ (1, 1)	Point(0.2)	B3
binbf01 (direction)	Beta ₊ (1, 1)	Point(0.5)	B1-B9
binbf01 (direction)	Beta ₊ (1, 1)	Point(0.55)	B9
binbf01 (direction)	Beta ₊ (1, 1)	Point(0.6)	B2, B5
binbf01 (direction)	Beta ₊ (1, 1)	Point(0.75)	B4
binbf01 (point)	Beta(1, 1)	Point(0.2)	B3
binbf01 (point)	Beta(1, 1)	Point(0.5)	B1-B9
binbf01 (point)	Beta(1, 1)	Point(0.55)	B9
binbf01 (point)	Beta(1, 1)	Point(0.6)	B2, B5
binbf01 (point)	Beta(1, 1)	Point(0.75)	B4
binbf01 (point)	Beta(40, 60)	Point(0.5)	B1-B9
binbf01 (point)	Beta(5100, 4900)	Point(0.5)	B8-B9
binbf01 (point)	Beta(60, 40)	Point(0.5)	B1-B9

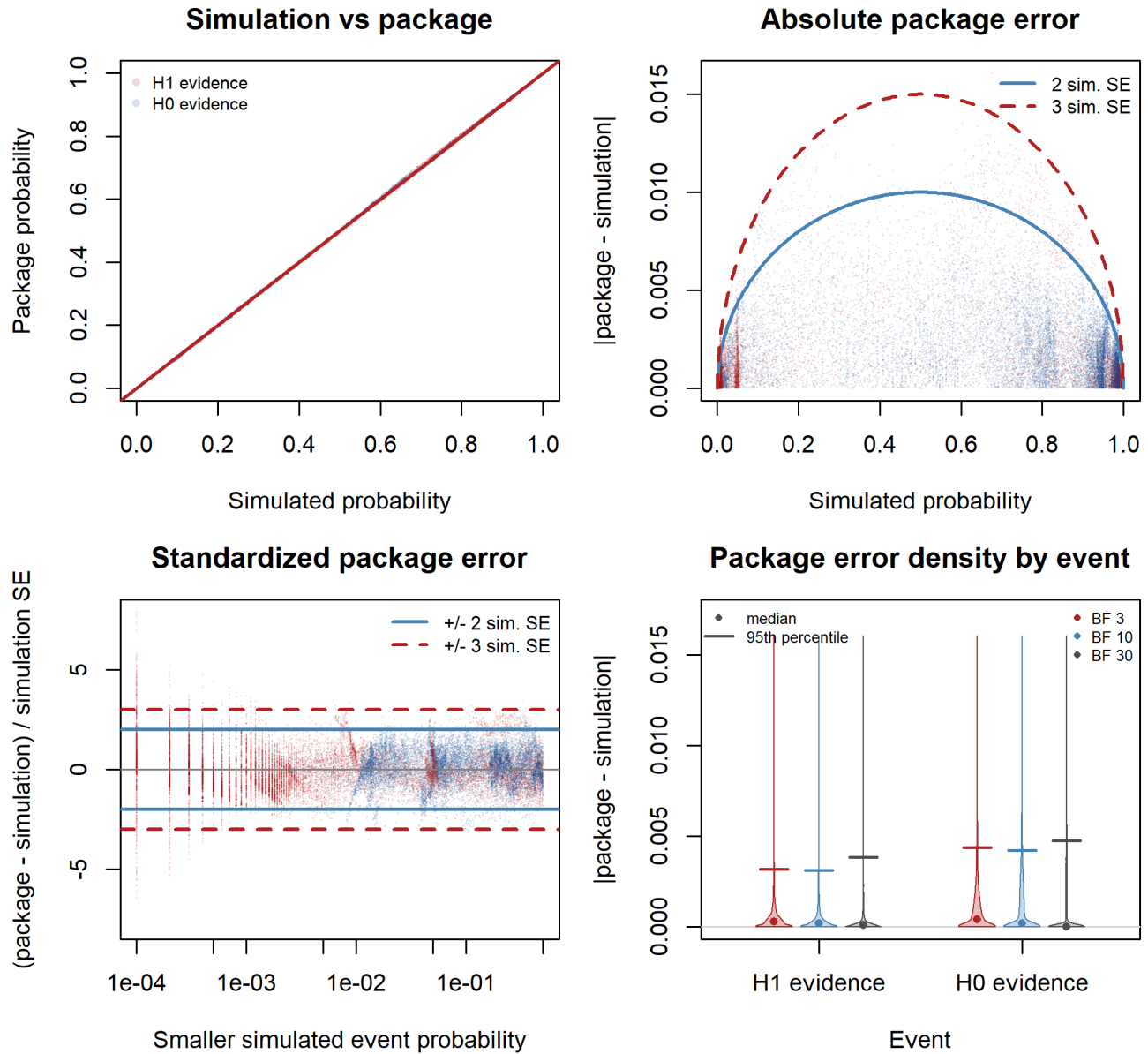
3.1 Fixed-Design

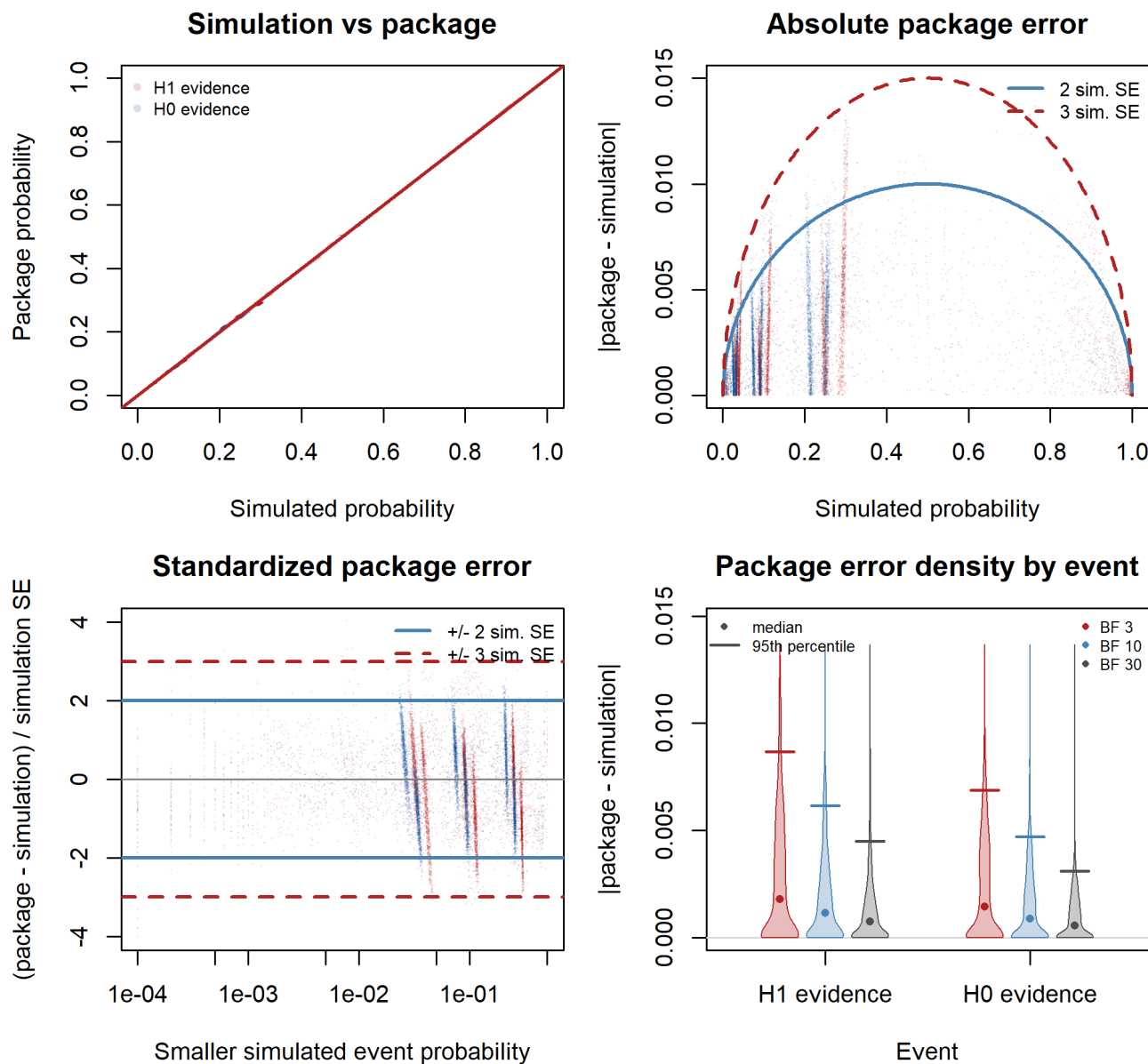
Table 11: Fixed-design fixture setups for Binomial Test analyses.

Function	Analysis priors	Evidence thresholds
binbf01 (point)	34	BF 3, 10, 30
binbf01 (direction)	14	BF 3, 10, 30

The fixed-design plots validate package probability calculations against the simulated event frequencies. Cached diagnostic rows are shown; for computationally expensive reference families, the plotted cache can be a stratified subset of the full fixture summary. The panels compare simulated and package probabilities, show absolute and standardized package error, and show package-error densities by event.

binbf01-point fixed-design fixtures



binbf01-direction fixed-design fixtures**3.2 Sequential-Design**

Sequential fixture schedules use looks every 10 observations: one family starts at $n = 20$ and includes 2–20 looks; the stress schedules start at $n = 10$ and include 50, 100, or 200 looks. The 100- and 200-look schedules are compatible with the longer design sequence only.

Table 12: Sequential-design fixture setups for Binomial Test analyses.

Function	Analysis priors	Evidence thresholds	Look sequence
binbf01 (point)	34	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)
binbf01 (direction)	14	BF 3, 10, 30	start 20, by 10 (2-20 looks); start 10, by 10 (50, 100, 200 looks)

Sequential binomial fixture settings are retained for future package-function validation. The package

does not currently expose a sequential binomial evaluation function, so no sequential binomial package-calculation figure is rendered.

4 Validation Summary

The tables and plots in this vignette are a compact summary of the fixture verification suite. They are intended to show which settings were checked, which simulations validate package probability calculations, and which settings are currently retained as integrity-checked fixtures for future function development. The automated package checks remain the authoritative pass/fail record for development.